

# Cloud-Fog Automation: Vision, Enabling Technologies, and Future Research Directions

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**Abstract**—The Industry 4.0 digital transformation envisages future industrial systems to be fully automated, including the control, upgrade, and configuration processes of a large number of heterogeneous wired/wireless interconnected devices in Industrial Internet of Things environments. Most of the industrial automation systems today are based on the traditional International Society of Automation (ISA)-95 model, with some recently transitioned to Cloud Automation systems. Latest developments in network connectivity technologies, artificial intelligence, and Cloud/Fog computing technologies have motivated us to rethink the ISA-95 model. In this article, we propose a vision that aims to migrate most of the computational and automation tasks closer to the ground, which we term the collaborative “Cloud-Fog Automation” paradigm. We perform a comprehensive survey of the state-of-the-art and formulate the three pillars of this vision: Deterministic connectivity, deterministic connected intelligence, and deterministic networked computing. In each of these pillars, we review their latency and reliability, security, and functional safety requirements and challenges. Finally, we articulate and highlight key future research directions to realize this vision.

**Index Terms**—Artificial intelligence, automation, cloud, Fog, Industrial Internet of Things (IIoT), Industry 4.0.

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## I. INTRODUCTION

THE Industry 4.0 digital transformation has seen innovations in industrial systems growing in leaps and bounds, which is largely propelled by significant advancements in Industrial Internet of Things (IIoT), cyber-physical systems (CPS), artificial intelligence (AI), machine learning (ML), Cloud computing, and data analytics capabilities for industrial operations [1]. The proverbial “alignment of stars” of these digital technologies have meant that many of the smart factory innovations today are able to provide efficient and responsive services to customers at a level not previously possible. These smart factory innovations also led to the rapid adoption of systems that provide predictive maintenance, Big Data analytics, self-optimization/self-healing, real-time visibility of manufacturing assets, and so forth, which can minimize unplanned downtime and in turn reduce revenue losses. This transformation is evident across many different industrial segments.

One crucial characteristic of Industry 4.0 is the ability of industrial systems to be fully automated, including automating the control, upgrade, and configuration of a large number of heterogeneous wired/wireless interconnected devices in harsh industrial environments. Ubiquitous networking in such environments is important to reduce maintenance, installation, and operation costs. Achieving full industrial automation has considerable challenges as networks need to cater for multi-dimensional requirements such as ultra-reliability, ultra-low latency, and secure communication over many thousands of nodes.

Research in industrial communication has sought to address these challenges. One important application of industrial communication is factory automation applications, involving discrete manufacturing processes typically regulated by feedback from sensors and actuators. These closed-loop systems have stringent performance requirements, especially in terms of bounded low-latency communication. Factory automation applications rely greatly on the determinism and capabilities of the underlying communication and networking protocols.

Despite concerted industrial efforts over the past two decades, there still exist major challenges in achieving the latency, reliability, and security requirements for mission-critical controls in factory automation applications. In addition to the communication and networking infrastructure, a large part of meeting performance requirements of industrial applications is dependent on the overall factory automation design paradigm.

There is a recent trend in the community for a paradigm shift in factory automation systems, moving progressively from a classical hierarchical pyramid architecture as defined by the International Society of Automation (ISA)-95 model [3], toward a more Cloud-based automation architecture, where upper layer applications such as Enterprise Resource Planning (ERP) and Manufacturing Execution Systems (MES) are deployed in the Cloud. The rapid developments of next-generation wireless technologies, and innovations in AI, IIoT, Cloud/Fog computing have motivated us to rethink the traditional ISA-95 model. We envisage a paradigm that deploys the lower layer applications such as supervisory controllers, machine controllers, and input and output (I/O) modules locally over wireless links at the Fog layer. The collaborative nature between the Cloud and Fog layers will play a key role in future factory automation systems. We propose this new architecture and term it the “Cloud-Fog Automation” paradigm.

The Cloud-Fog Automation vision was first conceptualized and introduced in [2]. In this article, we aim to further promote this new design paradigm for factory automation systems. By means of a survey article, we systematically interpret this vision, review the latest progress of the enabling technologies, and identify the technical challenges for future research. More specifically, we make the following contributions.

- 1) Propose a new Cloud-Fog Automation vision and outline requirements and challenges to realize this vision.
- 2) Survey and characterize key enabling technologies that form the three pillars of Cloud-Fog Automation: deterministic connectivity; deterministic connected intelligence; and deterministic networked computing, along with their performance metrics and technical requirements in latency and reliability, security, and functional safety.
- 3) Articulate and highlight key future research directions addressing the challenges in realizing the new Cloud-Fog Automation paradigm.

The rest of this article is organized as follows. Section II presents our Cloud-Fog Automation vision. Sections III–V review the state-of-the-art enabling technologies for deterministic connectivity, deterministic connected intelligence mechanisms empowered by AI, and deterministic networked computing technologies, respectively. Each of these sections summarizes the technical challenges in terms of latency and reliability, security, and functional safety. Section VI highlights research directions, and finally, Section VII concludes this article.

## II. CLOUD-FOG AUTOMATION VISION

In this section, we introduce the Cloud-Fog Automation vision, present the history of automation systems, and describe how the new paradigm can drive the next evolution of industrial automation systems for Industry 4.0.

### A. Evolution From a Layered Architecture to a Network-Centric Architecture

The adoption of automation systems in manufacturing has enabled the third revolution of industry since the 1970s [4].

Since the early 2000s, a pyramid automation architecture—the ISA-95 model [3] has matured in the design and integration of modern automation systems. In the ISA-95 model (left of Fig. 1), all the communication, computing, and control elements of the automation systems in manufacturing enterprises are both logically and physically partitioned into five separate levels (Levels 0 to 4), with each level corresponding to physical machines, low level machine control, middle level supervisor control, MES, and ERP. Since the 2010s, Industry 4.0 as driven by IIoT/CPS technologies, has dramatically changed the landscape of automation systems from the layered structure of ISA-95 into a more network-centric architecture [2]. In a network-centric architecture, an element at any of the five levels can be directly connected with another element at any of the levels according to the needs of the system, thanks to the pervasive connectivity of IoT. Therefore, the ISA-95 model has become more of a logical/virtual partitioning for semantic definition of the functionalities instead of the physical partitioning and topology of the elements.

### B. New Design Paradigm for Cloud-Fog Automation

Powered by the rapid penetration of the Cloud computing in professional and enterprise applications, deploying Level-3 and Level-4 functionalities of the ISA-95 model has become a common practice in the automation domain today, which we call Cloud Automation [center of Fig. 1, e.g., the NAMUR Open Architecture (NOA), Open Platform Communications Unified Architecture (OPC UA), Cloud Robotics]. As a further step, we foresee that Level-1 and Level-2 functionalities of the ISA-95 model can be migrated from the purpose-specific computing hardware such as programmable logic controller (PLC) and industrial PC (IPC) to the general-purpose and virtualized Fog computing infrastructure, which becomes the “Cloud-Fog Automation” paradigm we propose (right of Fig. 1). There are also major industry-driven initiatives toward this paradigm, such as the Open Process Automation Standard (O-PAS) driven by the Open Process Automation Forum (OPAF), OPC UA Field eXchange (OPC UA FX) standard driven by the OPC Foundation, Fog Robotics, and so forth.

In Cloud-Fog Automation, the Level-3 and Level-4 software functionalities of automation systems are hosted in the general-purpose Cloud computing infrastructure, and more importantly, the time-critical control functionalities at Level-1 and Level-2 are also hosted by general-purpose computing infrastructure which we call “Fog” in contrast with “Cloud.” Hence, the layered structure in ISA-95 pyramid becomes only logical, and the physical topology becomes much flatter, i.e., any node can access the information from any other node depending on the needs and security rules. The primary differentiation between Cloud Automation and Cloud-Fog Automation is that the Fog computing infrastructure applied in the latter can be fully independent from the Cloud computing infrastructure. Moreover, there should be sophisticated security and safety boundaries between the Fog and the Cloud. These boundaries should be certified against operational technology (OT) standards instead of information technology (IT) standards. Although Cloud providers

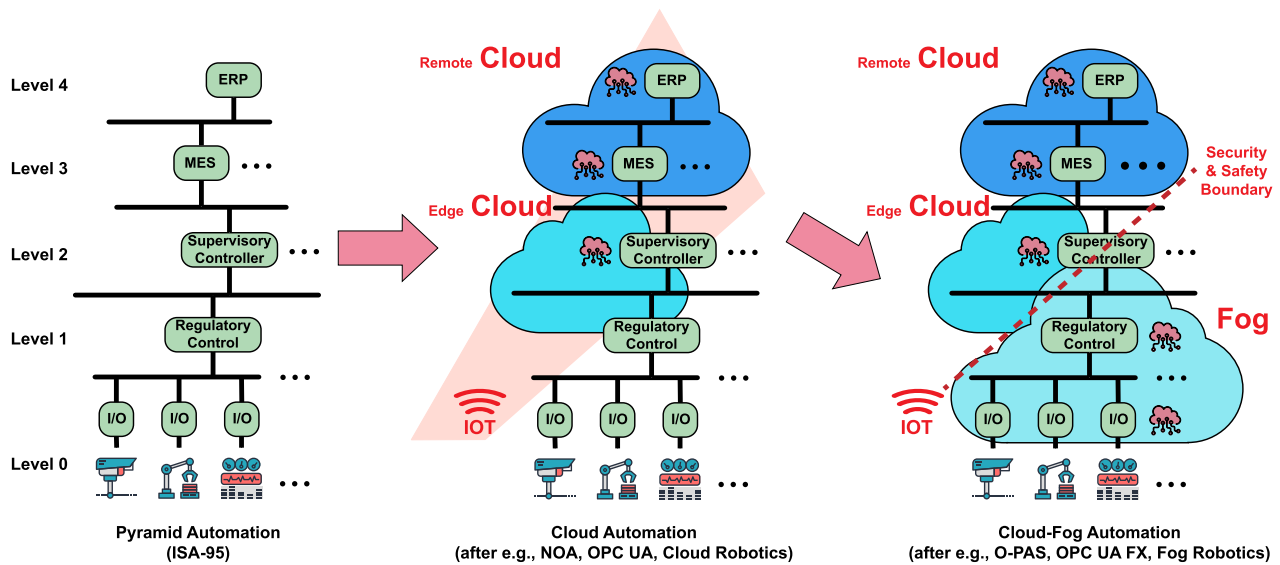


Fig. 1. Evolution of industrial automation systems design paradigms: From the pyramid automation of the ISA-95 model (left), to Cloud Automation (center), and the proposed Cloud-Fog Automation paradigm (right), based on [2].

often promise lower latency and stronger security by an “Edge Cloud” compared to a “Remote Cloud” (as illustrated in Fig. 1), an explicit boundary is still needed to better protect the Fog part. Although the terms Edge and Fog are often used interchangeably in literature, we emphasize the above differences in our specific context.

Another distinct feature of Cloud-Fog Automation is the replacement of wired networks at Level-1 and Level-2 with next-generation wireless technologies such as 5G Ultra-Reliable Low Latency Communications (URLLC), Wi-Fi 6/7, and so forth. It is important to note that the term “onloading” refers to the case where a computing task is loaded from a higher level to a lower level computing infrastructure, e.g., from Remote Cloud to Edge Cloud, from Edge Cloud to Fog, and from Fog to field I/O devices (if there is sufficient computing capability). This is an important feature to guarantee the availability of the overall system with respect to the stringent requirements of latency and reliability of Level-1 and Level-2 control, e.g., when the network or Cloud/Fog computing infrastructure is temporarily unavailable or under performing, the orchestration system will automatically move the time-critical tasks to closer to the data producer and consumer to reduce the negative consequences. In other words, Cloud-Fog Automation requires the two-way migration of computing tasks, in the upwards and downwards directions, depending on the application and users’ demand, and the situation of the infrastructure.

Compared with traditional Level-1 and Level-2 controllers, Cloud-Fog Automation promises much more powerful computing capabilities with lower costs, due to the rapid evolution of Fog and Edge computing technologies. The close-to-field and “physically distributed” nature of the Cloud-Fog Automation is the key to reducing latency, improving reliability and resolving security and safety concerns (note that Cloud and Fog are separated by a security and safety boundary, as illustrated in Fig. 1). The virtualization and “logically centralized” nature of

Cloud-Fog Automation will also significantly ease postcommissioning upgrading and reconfiguration, dynamic resource allocation and load balancing, onloading and offloading of modularized tasks, and digital distribution and delivery of products and services. Friendliness toward AI technologies also makes Cloud-Fog Automation a natural choice to realize emerging AI-powered automation systems that are difficult for traditional Level-1 and Level-2 controllers. When deployed over wireless networks, Cloud-Fog Automation promises to further reduce the costs of cabling and installations, hence enabling mobility, flexibility, and fine-grained deployments of automation systems.

From a business point-of-view, Cloud-Fog Automation will bring significant values to system integrators and their customers. It will provide more flexible deployments, reduce the Capital Expenditure (CAPEX) and Operational Expenditure (OPEX), increase hardware agnostic and vendor independency, and enable new use cases, such as mobility and AI-powered advanced control at Level-1 and Level-2. We can imagine a future where Cloud-Fog Automation can be provided/subscribed as a service as long as there is network connectivity. This will be enabled when emerging communication technologies are mature enough, e.g., when Wi-Fi 6/7, 5G/6G networks are deployed with multiaccess edge computing infrastructure. It will dramatically lower the barrier of using automation technologies for small businesses and developing markets, i.e., “affordable and inclusive automation” will become a reality through Cloud-Fog Automation. Consequently, automation systems will be increasingly provided as a service with varying degrees of new technical and infrastructure requirements, i.e., new business models of Control-as-a-Service, Robots-as-a-Service, and Manufacturing-as-a-Service in the near future. Robots-as-a-Service is currently an active area of research and development, and it has recently been provided on a service platform by *Ambi Robotics* [5]. We expect to see more momentum building in this area, and on a

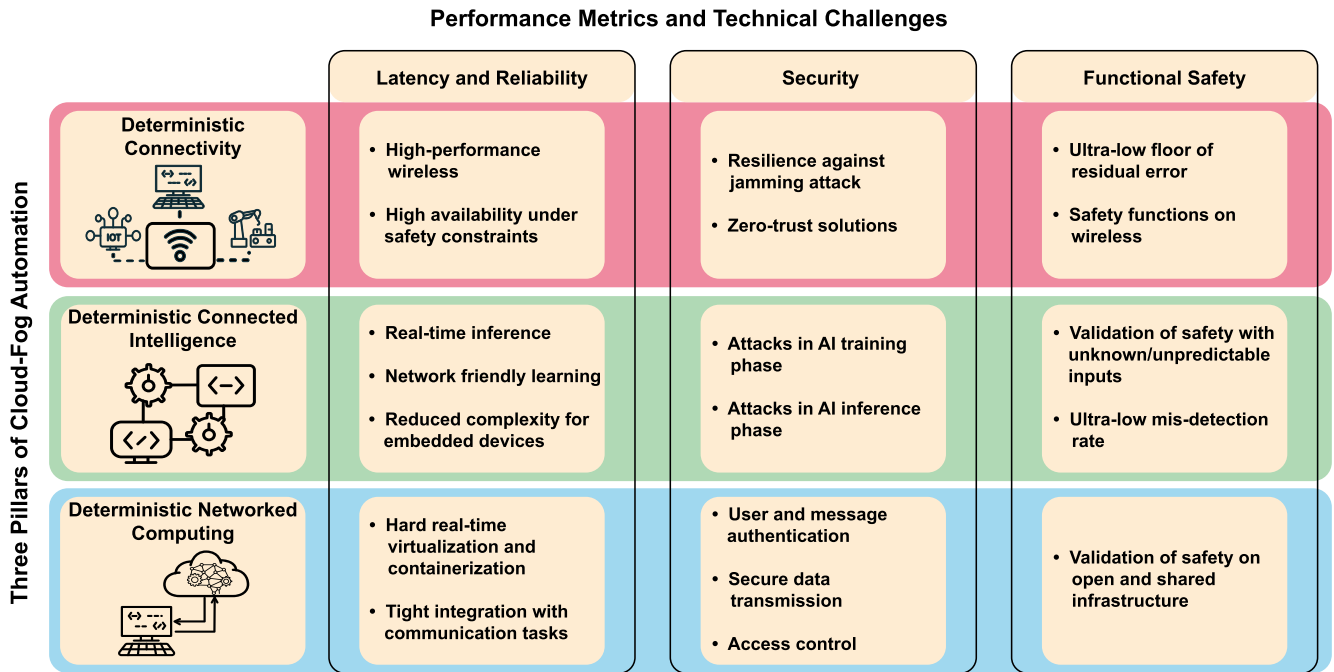


Fig. 2. Three pillars of key enabling technologies and the performance metrics/technical challenges in realizing the Cloud-Fog Automation vision.

similar basis, other services can also be realized with Cloud-Fog Automation.

### C. Enabling Technologies, Requirements, and Challenges

Building on established technologies in automation science and engineering, emerging technologies in the areas of *deterministic connectivity*; *deterministic connected intelligence*; and *deterministic networked computing* will be pivotal in the development of the Cloud-Fog Automation paradigm. *Deterministic connectivity* ensures the timeliness of data transmission to meet the stringent hard real-time requirements in IIoT. *Deterministic connected intelligence* deploys AI technologies (including the processes required to drive explainable AI, model training, real-time inference, etc.) in such a way that determinism is ensured and supported by the underlying deterministic connectivity technologies. *Deterministic networked computing* ensures that the computing and execution of the tasks are performed within the specified time frame within the networked IIoT environment. The coupling and interlocking of these three enabling pillars of technologies will be much tighter than ever. These pillars will enable the proposed paradigm to achieve the stringent requirements in *latency and reliability*; *security*; and *functional safety*. The benefits of the Cloud-Fog Automation should be delivered with comparable (if not better) performance to traditional solutions in these three dimensions. In Fig. 2, we summarize the key enabling technologies in the three aforementioned areas and the three dimensions of technical challenges. The rest of this article will delve into each of these aspects.

Interoperability of existing/newly proposed solutions and standards is crucial in all three pillars for the practical adoption

of the enabling technologies. For instance, it is important that all wired and wireless communication technologies are interoperable in a factory environment to achieve determinism in connectivity. To achieve deterministic connected intelligence, the programming languages, model training, and inference processes of AI systems need to be interoperable. Computing frameworks and systems (e.g., container/Docker-based and virtual machine-based computing) have to be interoperable to achieve deterministic networked computing.

## III. DETERMINISTIC CONNECTIVITY

In this section, we review network connectivity technologies for deterministic communication, which forms the first pillar of the proposed Cloud-Fog Automation paradigm.

### A. Latency and Reliability

1) *High-Performance Wireless*: The new Cloud-Fog Automation paradigm will replace wired connections at Levels 1 and 2 (of the ISA-95 model) with wireless connections. Wireless technologies will improve the mobility and flexibility of the industrial applications, hence driving new IIoT use cases and deployments. However, it is extremely challenging to provide the level of reliability, latency, security, and functional safety performance as provided by wired connections, due to inherent nature of the wireless medium and harsh industrial environments. Significant efforts have been invested recently to develop high-performance wireless solutions in industrial environments.

Industrial control typically consists of open-loop and closed-loop controls. The demand for high reliability and deterministic connectivity are significantly more stringent in closed-loop control industrial applications. Traffic patterns also dictate the



communication requirements. For example, applications that generate cyclical traffic with short time frames have a higher requirement for deterministic connectivity. However, the majority of wireless technologies used in IIoT are developed independently, without adequate considerations of industry-specific requirements. Therefore, it is imperative for the Cloud-Fog Automation paradigm to incorporate optimal high-performance wireless solutions that cater for industry-specific use cases.

Many of the studies currently focused on the improvement of IEEE 802.11 Wireless LANs (WLANs). WLAN is a popular choice due to its high capacity and availability in off-the-shelf products. Recent amendments to IEEE 802.11 have demonstrated performance (in soft real-time industrial applications) comparable to real-time Ethernet networks. The scheduling in 802.11ax/Wi-Fi 6 enables more efficient use of the available bandwidth, allowing multiple devices to transmit and receive data simultaneously, or reducing interference between adjacent networks [6]. 802.11be, also known as Wi-Fi 7, is currently under development and is expected to provide significant improvements. One of the key features being developed in 802.11be is multilink operation (MLO) with bandwidth aggregation. MLO is designed to enable a single device to utilize multiple frequency bands simultaneously to improve throughput and reduce latency. Other Quality-of-Service (QoS) enhancements related to latency in 802.11be is improved congestion control, which helps to prevent network congestion and ensure that all devices receive fair access to the available bandwidth. However, the underlying MAC layer (which uses enhanced distributed channel access) still fails to provide hard real-time guarantees on data delivery, making it challenging for IEEE 802.11 to be used in mission-critical applications, especially in the presence of interference.

Several studies have explored the modifications of WLAN's MAC layer. For example, nonorthogonal multiple access (NOMA) with IEEE 802.11n is proposed in [7] to meet industrial requirements. Contention-free MAC represents the most straightforward way to achieve deterministic communication. This typically includes the use of time-division multiple access (TDMA). In [8], a TDMA-based protocol (called RT-Wi-Fi) is developed to improve determinism in channel access and packet delivery. Two other TDMA-based techniques, DeT-WiFi [9] and Soft-TDMAC [10], are proposed to support multihop network topologies. WIA-FA [11] is proposed as part of the IEC 62948 standard for factory automation. WIA-FA's physical layer is based on IEEE 802.11, while its MAC layer is based on TDMA. It offers three communication modes and five levels of data priority for different sensing and control applications. Industry-specific techniques, such as Siemens' Industrial WLAN (IWLAN) uses PROFINET and Ethernet/IP over WLAN. IWLAN is currently being deployed in many industrial scenarios, such as in automatic guided vehicles (AGV) and cranes applications.

5G New Radio (NR) is a new radio air interface for 5G mobile communications from 5G Release 15, and is promising technology that aims to deliver URLLC with 1 ms user plane latency and 99.9999% successful packet transmission rate in unlicensed (NR-U) and licensed (NR-FDD, NR-TDD) spectrum (as defined in 5G Release 16 and 17). Many of the studies have compared

the reliability and latency performance of 5G and Wi-Fi 6 [12]. 5G NR is advantageous for licensed deployments where the spectrum is often scarcer. It is more challenging to achieve the promised URLLC in an unlicensed spectrum. In [13], an optimization problem (considering reliability restrictions with industry channel characterization) is formulated, and a method for evaluating 5G NR's reliable minimum cycle time is the proposed. It successfully achieved minimal (few milliseconds) cycle time and  $10^{-6}$  packet error rate (PER). Techniques involving PHY-MAC cross-layer performance optimization, with semipersistent scheduling at the MAC layer, and NOMA at the PHY layer are studied in [14].

Some key technologies in 5G also aim to provide deterministic connectivity. For instance, URLLC is a set of features in 5G networks that provide ultralow latency, high reliability, and high availability for mission-critical applications. In URLLC, the end-to-end latency can be as low as 1 ms. Furthermore, 5G networks can support network slicing, which allows operators to create virtual network slices that are customized to provide specific levels of performance, including latency and reliability. 5G networks can support time-sensitive networking (TSN), a set of standards that provide time synchronization and deterministic communication capabilities for industrial automation and other applications, which will be further discussed in Section VI. The state-of-the-art of TSN currently focuses on using wired connectivity to achieve determinism and real-time data transmission to satisfy the stringent QoS of industrial applications.

Apart from IEEE 802.11 and 5G NR, there are also some studies on achieving deterministic connectivity using other mechanisms. For example, WirelessHP [15] defines a new protocol stack at both PHY and MAC layers. At the PHY layer, it minimizes the impact of the PHY preamble and optimally selects the symbol time and the number of symbols. At the MAC layer, it uses TDMA with an accurate synchronization. Results comparable to EtherCAT have been achieved, positioning WirelessHP as a suitable alternative. In [16], the GALLOP system is designed to provide deterministic connectivity for wireless closed-loop control in multihop networks. Results have shown that the packet delivery ratio ranges from 95% to 99.9% depending on the network topology. A MAC protocol named "bandjacking" is proposed in [17]. This scheme aims to reduce the impact of other interfering wireless networks in the same license-free bands.

**2) High Availability Under Safety Constraints:** There has been an increasing demand for high availability in industrial automation systems. A system's availability has a direct impact on business revenue. Unplanned downtime can lead to significant revenue losses. It is also important that high availability is ensured under safety constraints. For example, typical safety-critical industrial automation applications require more than 99.9999% availability (as defined in IEC 61508). This can be achieved by either ensuring availability first, and then, address the safety issue globally or replicating safe building blocks to ensure availability [18].

Generally, there are two methods for increasing the system availability. The first is to increase the system's reliability, as shown in Fig. 3. High reliability is an essential component

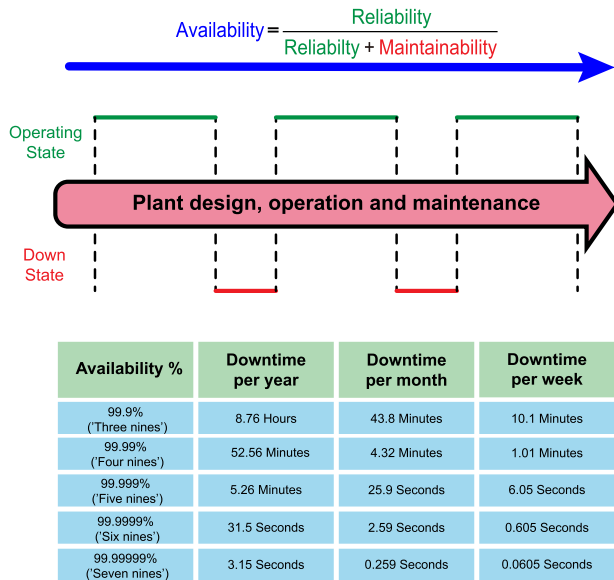


Fig. 3. Relationship between availability, reliability, maintainability, and downtime [19].

of maximizing availability in an industrial network. However, the corresponding reliability requirement for achieving a certain level of availability has not been fully considered. The second method is to implement redundancy or diversity. For instance, industrial environments and network topologies are usually dynamic. The availability of wireless industrial systems may suffer from any major changes in their environments or topologies. An approach to coping with the dramatic changes involved increasing the redundancy, such as multipath routing, high-availability seamless redundancy (HSR), and parallel redundancy protocol (PRP) [20]. The primary drawback of this approach is the significantly higher overhead, which in turn, results in significantly higher costs. Another option (without considering redundancy) is to use reinforcement learning to dynamically adjust and optimize network settings based on the collected network information [21].

## B. Security

1) *Resilience Against Jamming Attacks*: Security is particularly important when the use of wireless technologies begin to proliferate. Some security threats such as Probing Free Attacks can only occur in wireless networks. Although industrial wireless networks can be largely shielded from the outside world, attacks from neighboring warehouses (due to the leakage of electromagnetic fields) can still happen. The use of electromagnetic shielding can mitigate this attack, but it comes at high deployment costs. Different layers of the new Cloud-Fog Automation architecture are subject to different types of attacks, particularly the following:

- 1) internal attacks, where the attacker hijacks a legitimate node and gains access to sensitive information, and engage in further attacks, e.g., clone attacks and Sybil attacks;

- 2) external attacks, where the attacker can launch other wireless attacks near the area of the network deployment, e.g., jamming, spoofing, and eavesdropping.

Jamming attacks happen when jammers transmit jamming signals over legitimate channels. These attacks can be prevented with frequency/channel hopping, such as those applied to WirelessHART and ISA100.11a [22], [23]. However, if attackers target multiple channels simultaneously, then frequency/channel hopping cannot mitigate the attacks. Power control is another effective approach to counteract jamming attacks. A game theory-based power control framework is proposed in [24] and a joint power control/scheduling method in jammed networks is studied in [25]. A flexible antijamming strategy addressing relay selection and beamforming design problem is studied in [26]. In [27], intelligent reflecting surface (IRS) is argued to enhance antijamming effectiveness in wireless systems. IRS enhances signals of legitimate nodes while reducing interference caused by jammers. Several deep reinforcement learning (DRL) algorithms are studied in [28]. DRL can provide transmitters with an optimal communication policy without knowledge of the jammers' characteristics.

2) *Zero-Trust Solutions*: Cryptographic-based methods are traditionally use to mitigate various attacks. However, these methods are challenging to implement in industrial environments, where short messages are frequently exchanged. Zero-trust is a recent concept in cybersecurity, where no entity is to be trusted and the authentication system has to verify every entity requiring access to the system. Zero-trust has proved to be more feasible in industrial environments. Physical layer security (PLS) can suitably assist zero-trust solutions since it offers low computing complexity with excellent spatial differentiability and analytical unbreakability. PLS techniques for industrial control systems have demonstrated channel diversity, with enriched fingerprints in each channel, and also typically include three types of physical layer authentication (PLA) techniques that ensure the legitimacy of the transmitting nodes. The PLA-based approach for detecting clone and Sybil attacks has also been proposed. This method derives a fingerprint from the channel information between wireless nodes, and use ML algorithms to detect malicious attacks. Detection accuracy is improved by providing offline training samples with labels to the ML algorithm. An active PLA can be implemented without pilot signals or channel estimation, rather it can be deployed using an authentication signal embedding scheme [29].

## C. Functional Safety

1) *Ultralow Floor of Residual Error*: Functional safety in industrial environments depends on a system's safety function, i.e., how a system reacts to the input signals, and constantly ensure safety is maintained. Typically, the system will first sense a potentially unsafe state, process the data, and decide whether the entire system needs to enter a predefined safe state. The transition from an unsafe to a safe state should occur very quickly to prevent unnecessary safety incidents. Communication between safety devices and controllers need to be timely and reliable, which may be problematic in wireless networks.

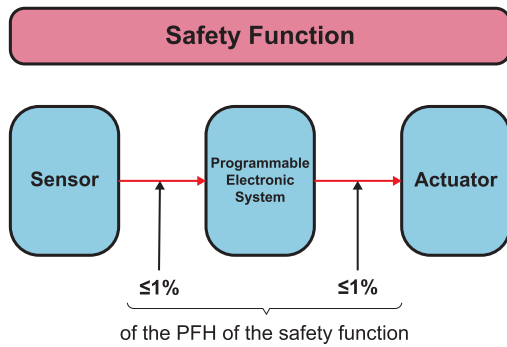


Fig. 4. Safety communication forms part of a safety function [31].

IEC 61508 is a functional safety standard that defines the life cycle model for electrical, electronic, programmable electronic, safety-related systems (E/E/PE). A safety integrity level (SIL) is specified in the standard. An SIL is the upper limit of the probability of system failure as controlled by the E/E/PE system. Failure probability is quantified using the probability of failures per hour (PFH). The allowed PFH for SIL 1 through SIL 3 varies from  $10^{-5}/h$  to a minimum of  $10^{-7}/h$ . It is recommended that each logical connection between the subsystems of the safety communication channels consumes at most 1% of the maximum PFH of the SIL (as shown in Fig. 4). Since PFH is based on the residual error rate (RER), ultralow reliability is required to achieve a ultralow RER. In [30], an analytical approach is used to calculate the RER probability distribution of fading channels. The approach is illustrated using two different cyclic redundancy check (CRC) polynomials corresponding to appropriate codeword lengths and safety data rates. The results showed that an appropriate CRC is crucial and that these CRCs, when used with filtering techniques, can achieve performance that is within the SIL.

**2) Safety Functions on Wireless:** Many of the approaches directly applied safety functions to wireless industrial networks, as described in IEC 61784-3-3 (PROFIsafe). An industrial safety communication based on IEC 61784-3-3, PROFINET, and WirelessHART is presented in [32]. A model based on IEEE 802.15.4e is defined in [33]. It estimates the safety function response time (SFRT) for systems that incorporate multiple inputs and multiple outputs. A gas detection system based on PROFIsafe over PROFINET and ISA100.11a is proposed in [34]. However, it is challenging to directly implement safety functions in wireless industrial networks. IEC 61784-3-3 specifies that safety controllers and devices should communicate cyclically through a time-triggered polling mechanism. Safety controllers initiate the communication by sending a data packet. This data packet should be acknowledged by the field instrument within the SFRT. A polling operation with a shorter cycle period can detect risks quicker. This is effective in one-to-one wired connections, but can be inefficient in resource-constrained industrial wireless networks as frequent cyclic polling can consume much of the bandwidth resources. A system guaranteeing the transmission of safety data is proposed in [35]. The system uses both software and hardware resources to ensure the

reliability and timeliness of data transmission in safety applications.

#### IV. DETERMINISTIC CONNECTED INTELLIGENCE

In this section, we review AI-based techniques to achieve deterministic connected intelligence, which will form the second pillar of the proposed Cloud-Fog Automation paradigm.

##### A. Latency and Reliability

**1) Real-Time Inference:** AI methods, such as deep neural networks (DNN), adversarial training, and transfer learning, have been widely used in various process diagnosis and prediction. Real-time inference is critical in ensuring its success. For example, defects monitoring has significantly improved when using neural network real-time inference approaches [36]. Instead of relying on inspectors, a convolutional neural networks (CNN) transfer learning method is proposed in [37]. This method implements automatic defect classification and significantly reduces the costs in smart manufacturing [37]. In [38], a deep CNN approach is combined an adaptive feature fusion with automatic continuous prediction to achieve higher predicting accuracy. Region-based fully convolutional networks can assist in guaranteeing a highly effective machine sorting in smart factories [39].

A dynamic feature selection technique based on genetic algorithm and artificial neural network (ANN) is proposed in [40]. This technique can recognize nonlinear relationships among the detected features and extract complex patterns from data collected by industrial sensors. A Gaussian–Bernoulli restricted Boltzmann machines (GBRBM) DNN algorithm is presented in [41]. This algorithm uses pretrained parameters from a multi-block GBRBM model and convert the fault detection in industrial environments into a classification problem. In the additive manufacturing prediction domain, CNN and ANN have been developed to analyze thermal images and process relevant process/design parameters separately [42]. Two CNN techniques for manufacturing are introduced in [43]. These techniques can improve the detection speed and accurately identify the locations of defects.

**2) Network Friendly Learning:** Network friendliness is important for deep learning (DL)-based automation in industry environments with Edge/Fog devices. Network friendly learning methods can improve the communication performance, and properly utilize the available resources. A method combining signal processing and deep residual learning model is proposed in [44]. Fog computing is an important aspect of industrial control in time-sensitive and mission-critical applications. A batched sparse scheme can reduce the computational and the communication overhead of the system.

Traffic congestion is a serious issue in manufacturing environments. Unbalanced resource allocation between the partitioned resources and computing efficiency on Fog-level devices should also be avoided. An inspection model with fast response time and low traffic load is proposed in [45]. In [46], Zipf's law is used to model the relationship between popularity ranks and the incurred computing costs in complex industrial environments.

It is used to calculate the popularity rank of IIoT services and predict the associated computing costs at the Fog level. Another approach is to use a context-aware application [47], which uses a placement policy to appropriately place Industry 4.0-oriented applications to ensure minimal service delivery time. A dynamic optimization scheme is proposed in [48]. This scheme dynamically coordinates and allocates the available radio resources based on the computational demands of the mobile nodes.

**3) Reduced Complexity for Embedded Devices:** Fog nodes are typically resource-constrained embedded devices. It is important to ensure that computational costs are minimized without sacrificing the accuracy of the DL algorithms. In [49], an offloading approach for DNN-based intelligent applications is proposed. This approach dynamically allocates tasks between the Cloud and Fog, and identify tasks that can be moved to Mobile Edge Computing (MEC) servers. However, the Cloud is not always reliable, especially in industrial environments with stringent computational latency and reliability requirements. To address this challenge, a DRL-based algorithm is developed to determine the channel variations and the task arrival patterns [50].

In [51], an adaptive genetic algorithm is proposed to automatically search the best rank for tensor decomposition to avoid memory space exhausted by the training of the Industrial Internet of Video Things system. In [52], a split-shuffle-residual (SSR) model reduces computational costs combining both split/shuffle and residual structure, while maintaining a high accuracy in result classification. Another challenging yet important issue persists in automatic surface defect inspection (ASDI) where production quality and efficiency are significantly affected. Hence, a ResMask generative adversarial network framework is proposed for rapid detection of high resolution images [53].

In [54], a bit-adjustment and energy-efficient convolution module using an approximate divide-and-conquer multiplier is proposed. This module is designed for compact and low-power edge processor design to efficiently process data at the Edge. An edge-intelligent service placement algorithm is presented in [55]. This algorithm finds a globally optimal solution using the particle swarm optimization inspired by nature. It solves the discontinuity of services caused by mobile devices that move around different MEC service areas. Although DL models have been widely used to optimize the calculation in industrial edge devices, there still exists a problem of computing resource shortage. The solution proposed in [56] combines an early exit model approach with knowledge distillation to accelerate inference from additional side branch classifiers. Industrial surveillance has become an increasingly important issue in cyber-physical security in industrial environments.

## B. Security

**1) Attacks in AI Training Phase:** AI security threat agents essentially destroys the integrity of AI model decisions. The adversary usually has two ways to achieve this outcome. First, the threat agent can contaminate the training set, i.e., intentionally adding artificial samples during the model *training phase*, so as to reduce the accuracy of the model obtained under such a

training set. Second, the threat agent explores the shortcomings of the model itself, such as constructing adversarial samples to make the model misclassify in the *inference phase*. There are usually two types of AI attacks in the training phase:

- 1) Data poisoning attacks [57]: The attacker adds a small amount of carefully constructed poisoned data to the training set of the model so that the model cannot be used normally during the testing phase or assists the attacker to invade the model without destroying the model accuracy. The former is a target-less attack, which is exploited to destroy the model usability; the latter is a targeted attack, which focuses on destroying the model integrity.
- 2) Backdoor attacks [58]: The attacker adds a carefully constructed poisoned data set to the training set of the model so that the poisoned model classifies the data with the trigger of backdoor (selected by attacker) into the target category of the attacker. For other normal samples, the model is still classified correctly.

Both of these attacks have strong concealment—the poisoned model shows normal predictive ability for clean data, and only outputs incorrect results for the target data selected by the attacker. Attacks with such a property can cause huge potential threats, and can cause serious accidents in mission-critical scenarios.

**2) Attacks in AI Inference Phase:** There are also two different types of attacks in the inference phase.

- 1) Adversarial attacks [59]: This attack refers to the malicious behaviors of using adversarial samples to deceive the model. Adversarial samples refer to malicious input samples formed by deliberately adding subtle interference in the dataset. These samples are visually indistinguishable from normal samples, but can lead to misclassification of AI models.
- 2) Evasion attacks [60]: This is an attack method that submits forged information to the biometric system to pass identity verification, which is an attack on the decision integrity of the biometric system based on AI technology.

Adversarial attacks usually originate from the defects of the AI model itself. Once used by an attacker, it will cause serious security hazards. Evasion attacks also demonstrate strong offensiveness. AI-based biological (including fingerprints, facial features, voice prints, and eye prints) verification technologies have been widely used. Another example is the forging of voice print recognition system. The attacker can pass the verification by playing a prerecorded or artificially synthesized unlocking audio to the voice print unlocking system. These attacks can easily compromise industrial automation systems that has in-built AI-based models.

## C. Functional Safety

**1) Validation of Safety With Unknown/Unpredictable Inputs:** In [61], a technique called N-HyMn is proposed. This technique infers a probabilistic Hybrid Automata model of the CPS and uses time-series data from operation to simulate control variables, then compared with actual I/O data to validate the functional safety requirements of the I/O data. To validate the



accuracy of the data, a method based on multilayer ANN with the Levenberg–Marquardt back propagation training algorithm is proposed in [62] to accurately estimate the pressure variability in the braking system, which is an critical safety requirement in the automotive industry.

**2) *Ultralow Misdetection Rate:*** A unified verification method for functional safety based on Markov analysis and model checking is proposed in [63]. This method uses model checking to estimate the effects of hardware and software failures in industrial automation systems. AI-based approaches have also been adopted in the functional industrial safety design. A method presented in [64] combines AI-based optimization techniques, estimation of distribution algorithms, iterated local search, and ant colony optimization, with formal verification tools for implementing safety-critical software to meet the safety requirements without sacrificing system performance and availability.

## V. DETERMINISTIC NETWORKED COMPUTING

In this section, we review Cloud/Fog computing technologies that aims to achieve deterministic networked computing, which will form the third pillar of the proposed Cloud-Fog Automation paradigm.

### A. Latency and Reliability

**1) *Hard Real-Time Virtualization and Containerization:*** Edge/Fog computing tends to favor containers to virtual machines (VM) due to the lower deployment time and resource consumption, which can then ensure real-time services in industrial environments. In a multiple algorithm service model (MASM), VMs can be logically separated in Cloudlets, with each Cloudlet being equipped with heterogeneous algorithms to adapt to different computational complexities and data sizes [65]. A virtual network embedding scheme for industrial wireless networks virtualization is proposed in [66]. In this scheme, the software defined network controller intelligently discovers the environment, quantifies the latency requirements, and set the forwarding decisions in the participating nodes accordingly. The use of lightweight virtualization and containerization is becoming more prevalent in Edge computing [67]. In [68], a dynamic microservice placement is explored. The goal of this technique is to achieve an optimal end-to-end latency for Edge-Cloud collaboration in smart manufacturing. A container-based virtualized automation control unit (CvACU) that allows online dynamic configuration and quick environmental adaptation is proposed in [69]. Container migration technologies, such as cold, precopy, postcopy, and hybrid migrations, are implemented in several emerging use cases [70].

**2) *Tight Integration With Communication Tasks:*** Constructing a collaborative automation industrial network requires system and software perspectives [71]. The traditional IEEE 802.15.4e Low Latency Deterministic Network mode is not capable of addressing low latency and high reliable requirements in factory environments. A twofold solution based on the adaptive transmission scheme (ATS) and rainbow product

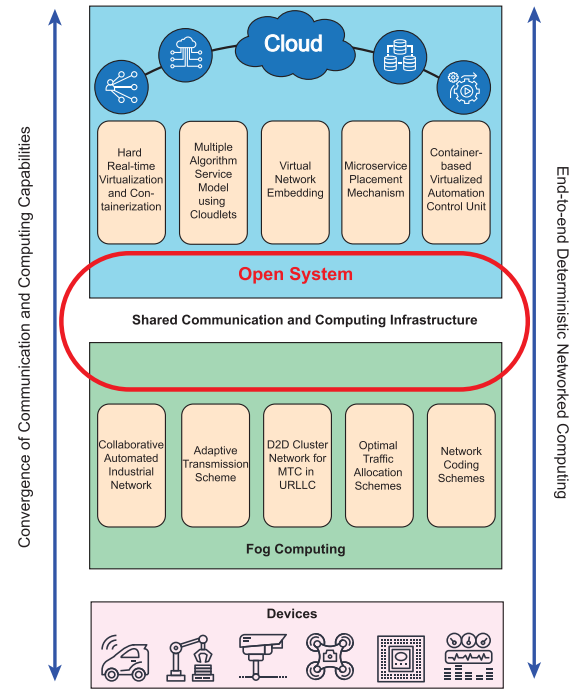


Fig. 5. Cloud-Fog paradigm for tight integration/convergence of communication and computing capabilities to achieve end-to-end deterministic networked computing.

ranking algorithm is developed in [72]. In [73], a tight integration is achieved by establishing a cluster of device-to-device networks. Devices within the cluster will dynamically assign tasks to neighboring devices/nodes to achieve machine type communication in URLLC. A specific factory automation use case is considered in [74]. In this use case, the central controller sends control messages to the robots and the actuating nodes in the network, leading to an effective collaboration and communication between the robots and the actuators. Network coding schemes, traffic allocation schemes, traffic dispersion, and network densification approaches have also been explored. Parallel sequence spread spectrum with code division multiple access can also be used to improve the speed of hardware implementation [75].

Fig. 5 summarizes the key technologies presented in this section for achieving end-to-end deterministic networked computing over an open system with shared communication and computing infrastructure.

### B. Security

**1) *User and Message Authentication:*** Traditional *user authentication* methods in Edge/Fog computing mainly rely on a public identification number (PIN) or a cryptographic private key (CPK) with a trust management center [76]. However, PIN-based methods are vulnerable to device clone attacks [77] and side-channel attacks [78]. The CPK-based scheme is not suitable for Edge computing when the centralized method meets the authentication requirement of large-scale of resource-constrained edge devices. A blockchain-based distributed authentication

scheme is proposed in [79], in which a Byzantine-based consensus algorithm is presented to realize activity traceability of Edge nodes, and the authentication data storage. An anonymous blockchain-based authentication scheme for smart grid systems is proposed in [80]. In [81], a blockchain-based mobility management system is proposed to reduce duplicate authentication between edge servers. Alternatively, there are several hardware-based authentication approaches that host applications securely [82]. These techniques extract unique features of in-built hardware of edge devices (e.g., the frequency response of microphone and/or loudspeaker) as fingerprints for authentication.

*Message authentication* is used to ensure the integrity and authenticity of the transmitted data in Edge computing. Digital signature authentication based on public key infrastructure (PKI) is the most common approach [83]. However, many of the PKI-based authentication methods still suffer from excessive computational and communication overhead. In [84], an efficient batch message authentication mechanism is proposed. This technique exploits digital signature techniques without certificates, which can be used to verify multiple messages simultaneously. In [85], a conditional privacy-preserving authentication solution using elliptic curve cryptography (ECC) is proposed. An anonymous message authentication scheme is proposed in [86] with the assumption that Edge servers are semitrusted entities. This technique uses proxy reencryption and group signature techniques to ensure the anonymity, confidentiality, and integrity of the data transmitted. Secure channel codes and universal hash functions can be leveraged to achieve message authentication with perfect security.

**2) Secure Data Transmission:** Secure data transmission is critical in preserving data confidentiality in Edge/Fog computing. Traditional methods typically use asymmetric encryption techniques (e.g., Diffie–Hellman) for key distribution, and symmetric encryption schemes (e.g., Data Encryption Standard and Advanced Encryption Standard) for data confidentiality. In [87], a signcryption scheme based on bilinear pairing for simultaneous data encryption and digital signature is developed. To improve the computational overhead, a lightweight and anonymous aggregate signcryption using ECC is proposed in [88]. Physical layer-based secure transmission methods can also be used for Edge computing in industrial environments. For example, a secure channel coding can replace the traditional channel coding at the physical layer to ensure transmission reliability and data confidentiality.

**3) Access Control:** Access control is used to authorize legitimate users/devices to access distinctive resources (e.g., privacy-sensitive data, bandwidth, and storage) in Edge/Fog computing [89]. In [90], a role-based access control scheme is proposed for Cloud storage, where the users' role are used for authorization of users. In [91], a identity-based proxy reencryption approach is proposed for leakage resilient access control in Fog computing. The cipher-text encrypted by data owner's public key is reencrypted through a proxy such that it can be decrypted by the data requestor's private key. Some Cloud and Edge collaborative schemes for fine-grained access control is presented in [92].

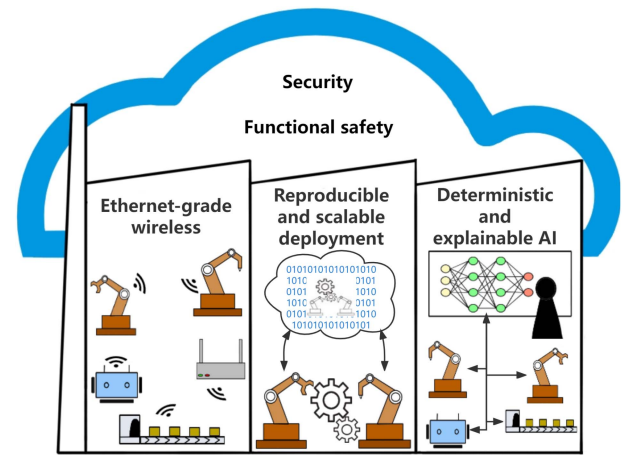


Fig. 6. Summary of highlighted research directions to fully realize the Cloud-Fog Automation vision.

### C. Functional Safety

Functional safety validation can be broached from the perspectives of both computing and communication infrastructure. From the perspective of the computing infrastructure, adopting safety-critical indicators and techniques should be consider to ensure functional safety in the CPS design [93]. The technique proposed in [94] utilizes temporary unobservability to analyze the soft error effects through Single Event Upsets at the gate-level in hardware. At the PLC level, a Boolean expression-based safety filter specifying functional safety requirements is proposed in [95]. From the perspective of the communication infrastructure, fault-tolerant control barrier functions can be used to avoid sensor faults that may cause safety violations [96]. The unique characteristics of distributed industrial control systems are explored in [97]. A colored petrinet approach is then developed to concurrently evaluate the impact of communication delay and the availability of distributed networked control systems for safety validation.

## VI. RESEARCH DIRECTIONS HIGHLIGHTS

In this section, we highlight five future research directions for driving the Cloud-Fog automation vision (as summarized in Fig. 6), not only filling current knowledge gaps but also serve as a call-to-arms to the industrial research community.

### A. Ethernet-Grade Wireless

Timely communication and precise mutual synchronization in sensing, computing, and control devices are critical in modern automation systems. The IEEE 802.1 Time Sensitive Networking (TSN) Task Group have developed a set of standards to support a range of time-, mission-, and safety-critical applications. TSN aims to achieve deterministic connectivity by modifying the Ethernet standard. These advancements will see more industrial applications migrating to wireless networks for higher mobility/flexibility and lower maintenance costs. The extension of TSN capabilities to wireless networks becomes the

next logical step. Reliable support of industrial applications over wireless networks will open up new markets, e.g., mobile robots, software-defined factories, and so forth.

Recent advances in IEEE 802.11 and 5G have demonstrated potential for TSN enhancements [98], [99]. There is currently considerable traction in integrating TSN with Wi-Fi [100], [101] and 5G [102], [103]. In 5G Release 16, it is possible to integrate the TSN through a fully centralized configuration model that supports scheduling individual connections using per-queue gating control (IEEE 802.1Qbv) and device synchronization based on IEEE 802.1AS (gPTP). Support for the TSN over 5G can be enabled by the several key functions, such as TSN synchronization via 5G bridge, 5G bridge and port TSN capability reporting and configuration, discovery of neighboring TSN topology on 5G bridges, reporting and management of neighboring topologies, and management of 5G data paths. As for Wi-Fi, the upcoming release 802.11be supports TSN through several enhancements such as the Scheduled Resource Protocol, enhanced QoS, accurate time synchronization, and improved channel access coordination. In [104], a hybrid TSN network proof-of-concept was developed to demonstrate that their proposed hybrid TSN architecture can maintain the essential TSN services.

However, delivering TSN-grade capabilities over wireless technologies remains a significant challenge and research in this area is currently still exploratory [6], [105]. Since TSN was originally designed for wired connections, many fundamental assumptions need to be reviewed and revisited. There are research gaps yet to be filled to realize TSN in the wireless domain [106], for example, the following.

- 1) TSN parameters should be configured based on the capabilities of wireless links. TSN link layer discovery and configuration protocols should be extended to ensure the management interface of TSN is wireless aware.
- 2) New parameters and abstraction interfaces between existing TSN functions and wireless protocols/standards need to be developed. A common abstraction interface for the underlying wired and wireless links will facilitate the deployment of the TSN over wireless.
- 3) New experimental and prototyping platforms, e.g., using Software Defined Radios are needed to validate research in practice.
- 4) New test plans including use cases, assumptions, and deployment scenarios need to be clearly documented and standardized in test plans to push for adoption.

The Internet Engineering Task Force has also standardized deterministic networking (DetNet) to achieve similar objectives as TSN. DetNet operates at Layer 3 of the Open Systems Interconnection Model stack, and defines the worst-case bounds on latency, low jitter, and ultrahigh reliability. Specific use cases for DetNet are limited since the standardization efforts are still ongoing. Future research should explore the integration of DetNet with wireless technologies, e.g., Reliable and Available Wireless [107] extends DetNet's architecture to address issues specific to wireless links. Challenges abound when network topologies dynamically change, and the allocation network

resources at run-time and the dynamic setup of new communication paths remain future work.

## B. Reproducible and Scalable Deployment

Future automation systems should focus on the convergence of communication and computing infrastructure. Despite being fully tested in laboratories/warehouses, wireless solutions often fall short in industrial scenarios upon deployment. The high variability of industrial settings require constant adjustment and tuning of wireless solutions. A wireless solution that is high performing in one scenario can be low performing in another scenario. Channel modeling is used to address this challenge. However, it is still challenging given the physical environmental variations in industrial sites. Ideally, there should be a generic channel model. This challenge can be addressed with digital twins. All physical objects are mirrored digitally in digital twins, hence the precise construction of digital twin wireless channels (by extracting wireless features in specific industrial scenarios) is possible [108].

Future research will include more efficient and scalable management of dynamic network topologies. Wireless industrial systems perform poorly when there are significant environmental and topological changes. Current approaches such as multipath routing and HSR increase redundancy, but incurs large overhead. New approaches using reinforcement learning to dynamically optimize network settings are much preferred, but yet to be extensively studied. To achieve IT and OT convergence, network function virtualization can be used to flexibly orchestrated into different network layers with in Edge or Fog devices. This can be applied to PLCs, data collection, aggregation, and analytic functions. IT-level technologies such as ERP, can be deployed in local/remote Cloud, and OT-level technologies such as MES, production domain, and cell controllers can be virtualized OT functions that are orchestrated flexibly depending on service requirements. Dynamic task offloading should also be considered in the automation framework.

## C. Deterministic and Explainable AI

The lost of human agency in the decision-making process of industrial automation tasks can have serious consequences. Deterministic and explainable AI (XAI) services are critical to drive AI-based automation. Previous studies have proposed various XAI-based techniques, such as model-driven deep deterministic policy gradient algorithm that explores optimal assembly policy, an automatic learning rate scheduler for the CNN-based fault classification, tuning parameters to guarantee deterministic system performance, and so forth [109], [110]. XAI-based techniques in harsh industrial environments are also explored, for example, the use of collaborative semantic inference, fuzzy logic for comprehensible deep networks, fuzzy Choquet Integral, and an explainer to diagnose model limitations [111]. XAI models and their parameters need to be tuned precisely to avoid/mitigate any potential risks. Given the involvement of human lives and assets of significant value, it is particularly important to understand how the deployed AI models operate. This leads to



the concept of interpretability of White-Box AI in industrial environments. Research in deterministic and XAI is still emerging, and their applications to industrial automation systems are relatively new, so there are significant opportunities for future research.

#### D. Security

In industrial environments, security risks and threats have expanded from the digital/cyber environment to the cyber-physical realm where threat agents can manipulate physical objects to compromise the system. For example, denial-of-service attacks, false data injection attacks, and routing attacks, can be launched to disrupt and interfere with the normal operation of industrial networks. The intrusion detection system (IDS) is one of the most effective approaches to protect factory assets and services from malicious activities and unauthorized usage. The IDS can monitor the operations of a host or a network, detect threats, and alert the system administrator when there is a security violation.

However, it is very challenging to devise an efficient IDS for industrial systems, partly due to the unique features such as distributed and dynamic systems, constrained resources, and heterogeneous devices. AI and ML-based techniques can be employed to dynamically learn traffic patterns and detect intrusion. Future research should focus on the periodicity of learning model updates to improve the performance. Conventional DL models have many parameters, hence requiring a large amount of computational resources, which makes it unsuitable for real-time detection in factory environments. Therefore, the development of lightweight DL and distributed learning models for the IDS should be explored as part of future work.

#### E. Functional Safety

Functional safety is critical in every stage (design, deployment, operation, and maintenance) of industrial automation systems. Current techniques primarily focused on the design and verification aspects of functional safety. Future systems will require more focus on the validation of functional safety deployed in dynamic operation and control processes. In selection of an optimal CRC is essential in designing a safe industrial communication system. Currently, the error detection capability of CRC in many protocols and standards are limited. For instance, CRC-16 is added for every 3824 bits or CRC-24 for up to 8448 bits for 5G NR. Therefore, a new design of CRC with strong error detection capability (e.g.,  $10^{-15}$ ) or a new error pattern model (without relying on CRC), is desired.

Future research directions should consider human-in-the-loop functional safety issues. AI-based techniques will also be increasingly adopted to predict and detecting functional safety parameters. AI will assist in optimally integrating various techniques into a unified system to ensure that functional safety at every level of the Cloud-Fog Automation paradigm. Although automation systems are increasingly moving toward machine intelligence, the presence of human is still essential. Functional safety techniques that revolves around human-in-the-loop is

limited (most current methods focused on human–robot Collaboration), and will need to be explored further.

## VII. CONCLUSION

Recent innovations and developments in network connectivity technologies, AI-based prediction and decision-making techniques, and Cloud/Fog networked computing have motivated us to rethink the traditional ISA-95 model widely used in industrial automation systems since its conceptualization in the 2000s. In this article, we proposed a vision that envisages a moving computational tasks and automation processes closer to factory environments, which we term the “Cloud-Fog Automation” paradigm. This vision is solidly grounded on the three pillars presented in this article, namely, deterministic connectivity, deterministic connected intelligence, and deterministic networked computing. We discussed how key enabling technologies in these three pillars can meet the stringent performance requirements and significant industrial challenges in terms of latency and reliability, security, and functional safety, thus realizing the Cloud-Fog Automation vision. We also outlined future research directions that can drive us toward a more localized automation infrastructure. It is our hope that this article will serve as both a survey of the state-of-the-art, and also a call-to-arms that motivates the industrial automation community toward a fully integrated, collaborative, and holistic Cloud-Fog Automation paradigm for Industry 4.0.

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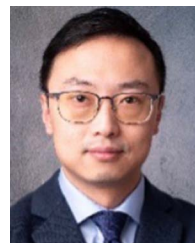


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